

Introduction to AWS Lambda

Lab overview

The lab provides a basic explanation of AWS Lambda. It will demonstrate the steps required to get started to create a Lambda function in an event-driven environment.

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources for you, making it easy to build applications that respond quickly to new information. AWS Lambda starts running your code within milliseconds of an event such as an image upload, in-app activity, website click, or output from a connected device. You can also use AWS Lambda to create new back-end services where compute resources are automatically triggered based on custom requests.

<https://prezi.com/p/qatfntzevzkx/aws-lambda-overview/>

Objectives

By the end of this lab you will be able to:

- Create an AWS Lambda function
- Configure an Amazon S3 bucket as a Lambda Event Source
- Trigger a Lambda function by uploading an object to Amazon S3
- Monitor AWS Lambda S3 functions through Amazon CloudWatch Log

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.

Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console**.

You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the **Region** unless instructed.

Lab environment

This lab demonstrates AWS Lambda by creating a serverless **image thumbnail application**.

The following diagram illustrates the application flow:

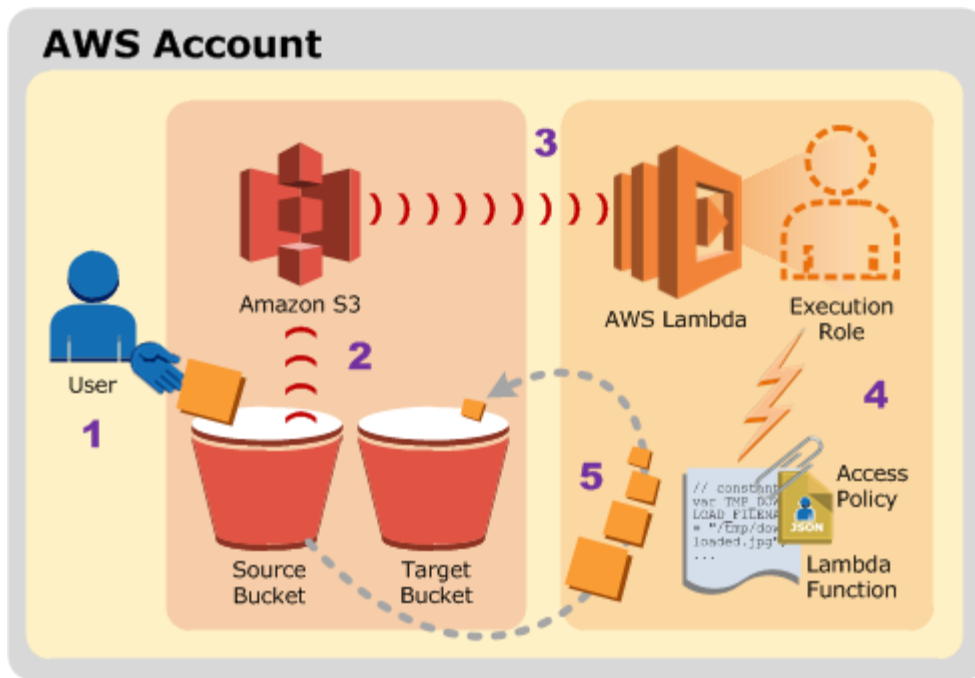


Image description: In the preceding diagram, the application flow is shown by the steps described below

- 1** A user uploads an object to the source bucket in **Amazon S3** (object-created event).
- 2** Amazon S3 detects the object-created event.
- 3** Amazon S3 publishes the object-created event to AWS Lambda by invoking the Lambda function and passing event data as a function parameter.
- 4** AWS Lambda executes the Lambda function.
- 5** From the event data it receives, the Lambda function knows the source bucket name and object key name. The Lambda function reads the object and creates a thumbnail using graphics libraries, then saves the thumbnail to the target bucket.

Upon completing this tutorial, you will have the following resources in your account:

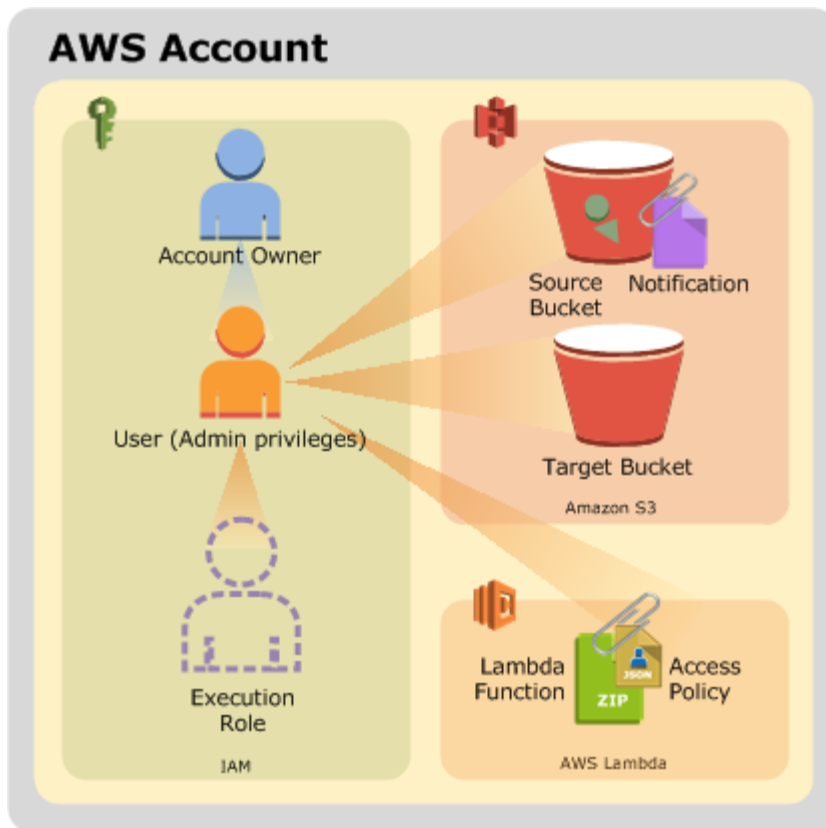


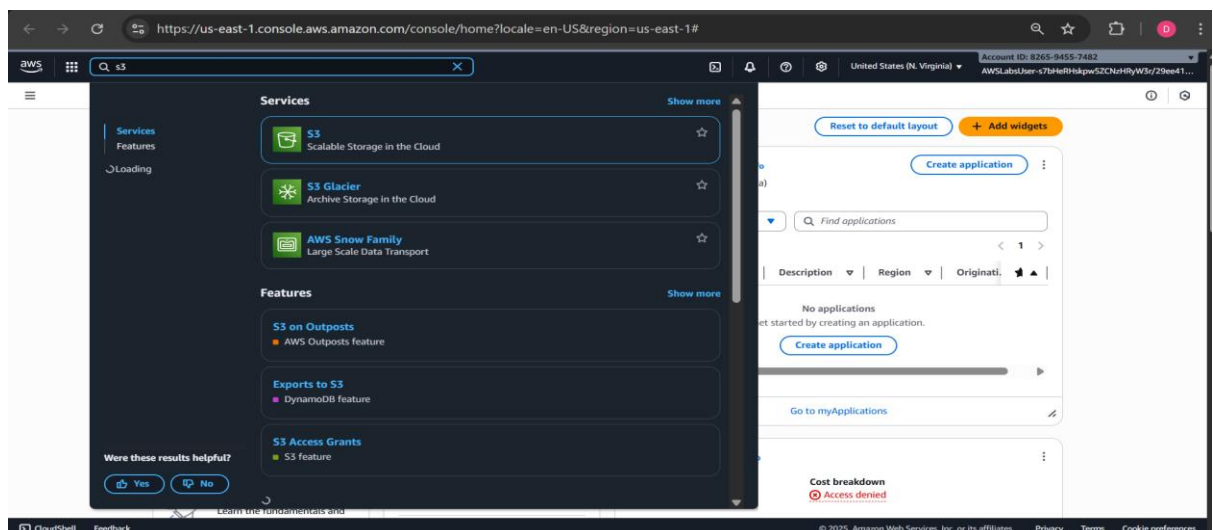
Image description: In the preceding digram, the user is shown having connections to both source and target S3 bucket, as well as access to use an AWS Lambda function through the given access policy

The steps in this lab show you how to create the Amazon S3 buckets and the Lambda function. You then test the service by uploading images for resizing.

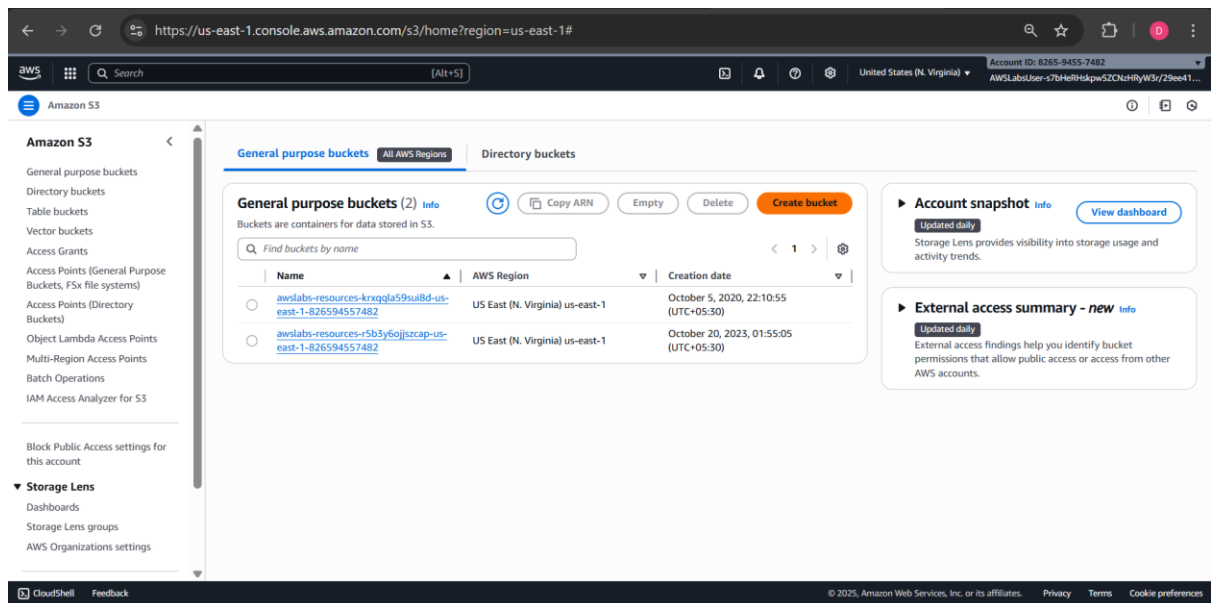
Task 1: Create the Amazon S3 buckets

In this task, you create two Amazon S3 buckets – one for input and one for output.

- At the top of the AWS Management Console, in the unified search bar, search for and choose S3

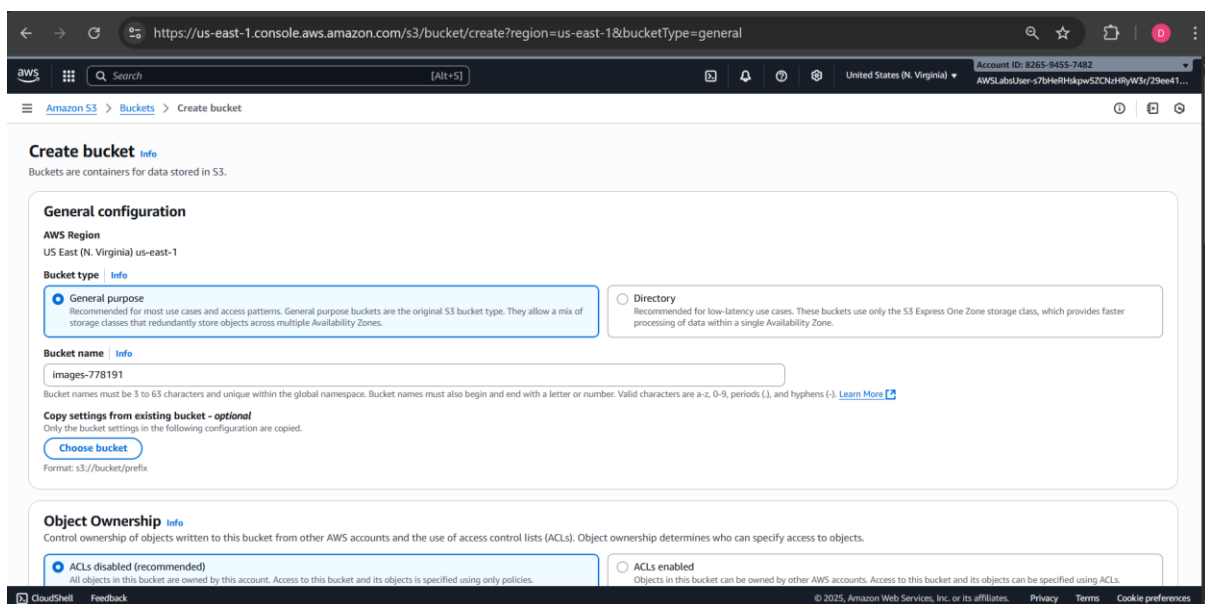


4. Choose **Create bucket**.



5. For **Bucket name**:, enter `images-randomnumber`, but replace **randomnumber** with an actual random number to ensure a unique bucket name such as `images-123456789`.

Note: Amazon S3 buckets require unique names regardless of region.



6. Copy the name of your bucket to a text editor for later use.
7. Keep everything else as default and choose **Create bucket** (at the bottom of the screen).

Note: If you receive an error stating **The requested bucket name is not available**, then choose the first **Edit** link, change the bucket name to a different random number and try again until it works.

8. Create another bucket for the output. Choose **Create bucket** and create another bucket with the same name as the first, but with `-resized` appended to the name of this new bucket, such as `images-123456789-resized`:

You should now have buckets named something similar to **images-123456789** and **images-123456789-resized**.

https://us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1&bucketType=general

Amazon S3 > Buckets > Create bucket

Create bucket

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)
images-778191-resized

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn More](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☒ **ACLs disabled (recommended)**
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ **ACLs enabled**
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

https://us-east-1.console.aws.amazon.com/s3/buckets?region=us-east-1&bucketType=general

Amazon S3 > Buckets

Successfully created bucket "images-778191-resized"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

General purpose buckets [All AWS Regions](#) [Directory buckets](#)

General purpose buckets (4) [Info](#)

Buckets are containers for data stored in S3.

[Find buckets by name](#)

Name	AWS Region	Creation date
aws-labs-resources-krxqqla59uul8d-us-east-1-826594557482	US East (N. Virginia) us-east-1	October 5, 2020, 22:10:55 (UTC+05:30)
aws-labs-resources-r5b3y6ojzcap-us-east-1-826594557482	US East (N. Virginia) us-east-1	October 20, 2023, 01:55:05 (UTC+05:30)
images-778191	US East (N. Virginia) us-east-1	September 16, 2025, 12:44:19 (UTC+05:30)
images-778191-resized	US East (N. Virginia) us-east-1	September 16, 2025, 12:45:26 (UTC+05:30)

Account snapshot [Info](#)
[View dashboard](#)
Updated daily
Storage Lens provides visibility into storage usage and activity trends.

External access summary - new [Info](#)
[Updated daily](#)
External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

9. Download the file **HappyFace.jpg** to your local machine (*Save link as...*).

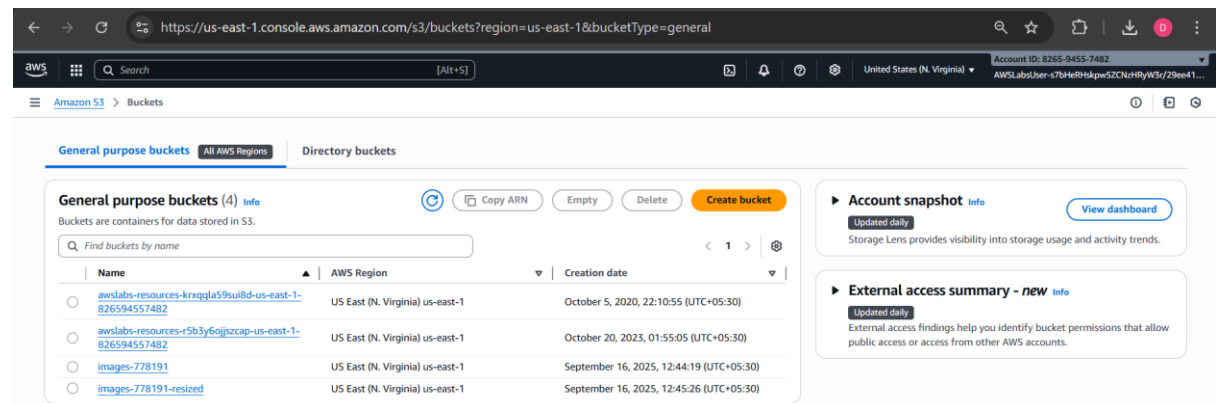
- [HappyFace.jpg](#)

10. Open the image on your computer.

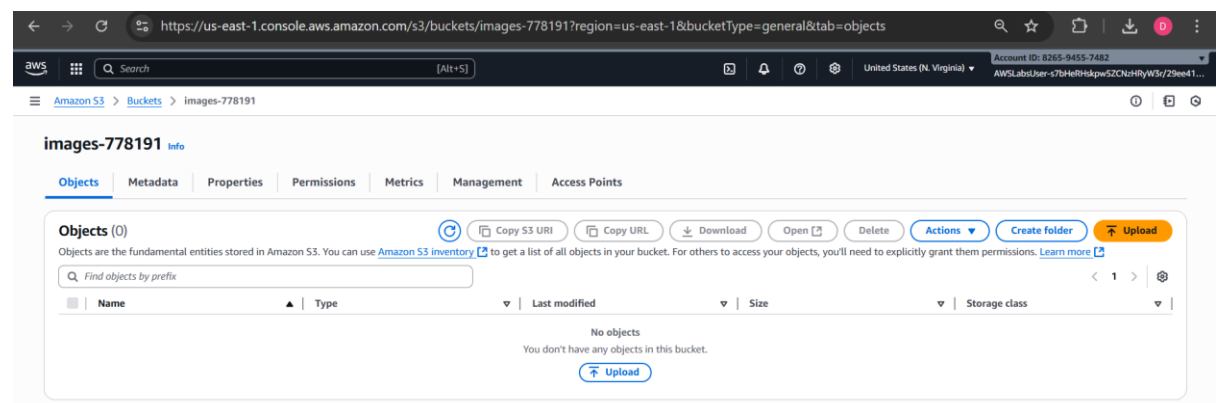
- It is a large picture, with dimensions of 1280 x 853.



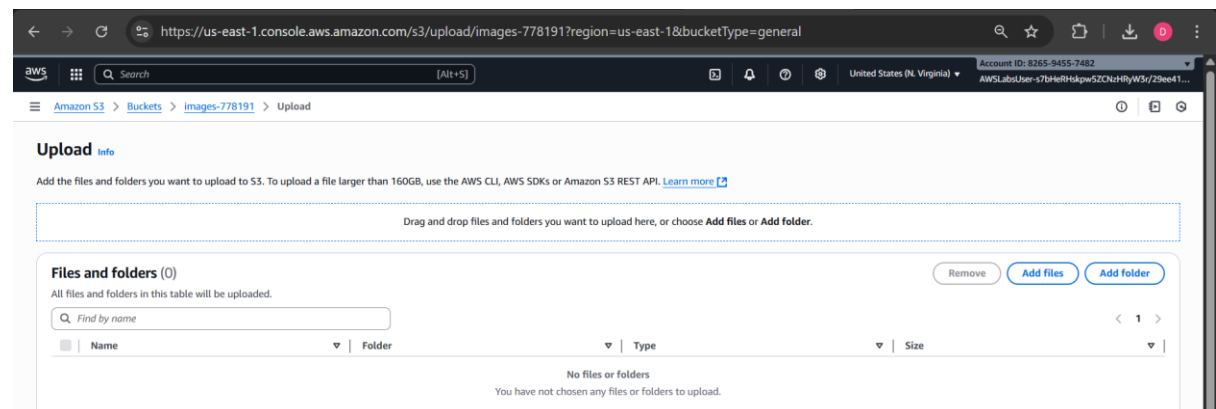
11. In the AWS console, choose the link to the first bucket you created (the one without the -*resized* appendix).



12. Choose **Upload**.



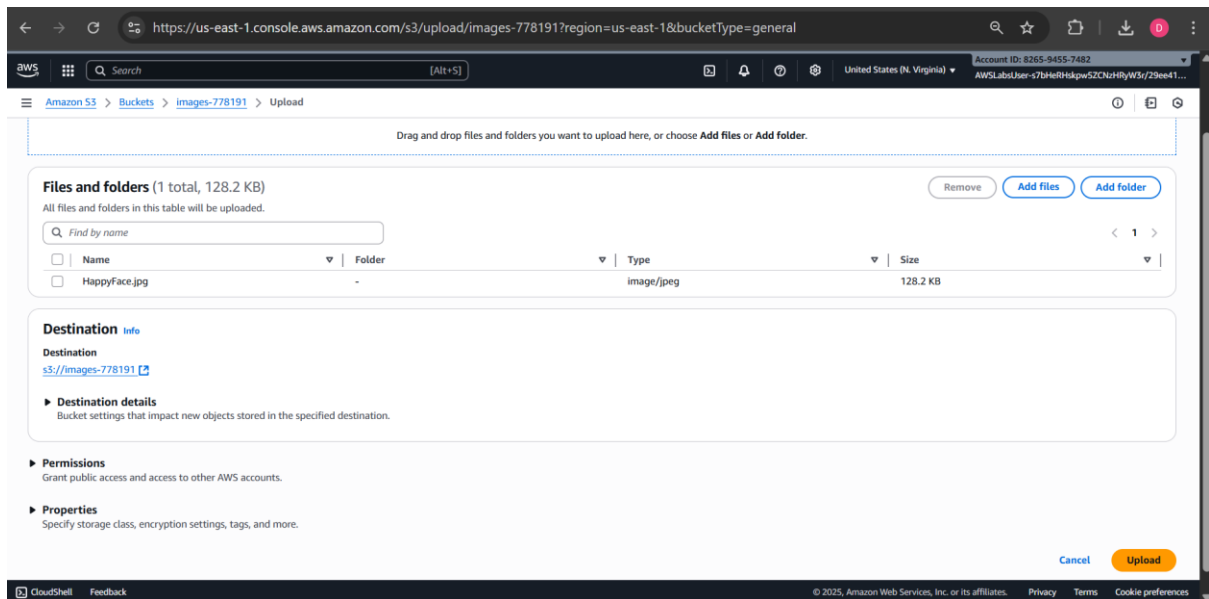
13. Choose **Add files**.



14. Use your file explorer and select the **HappyFace.jpg** file you just downloaded to your local machine.

15. Choose **Upload**.

Note: Later in this lab you invoke a Lambda function manually by passing sample event data to the function. The sample data refers to this *HappyFace.jpg* image.

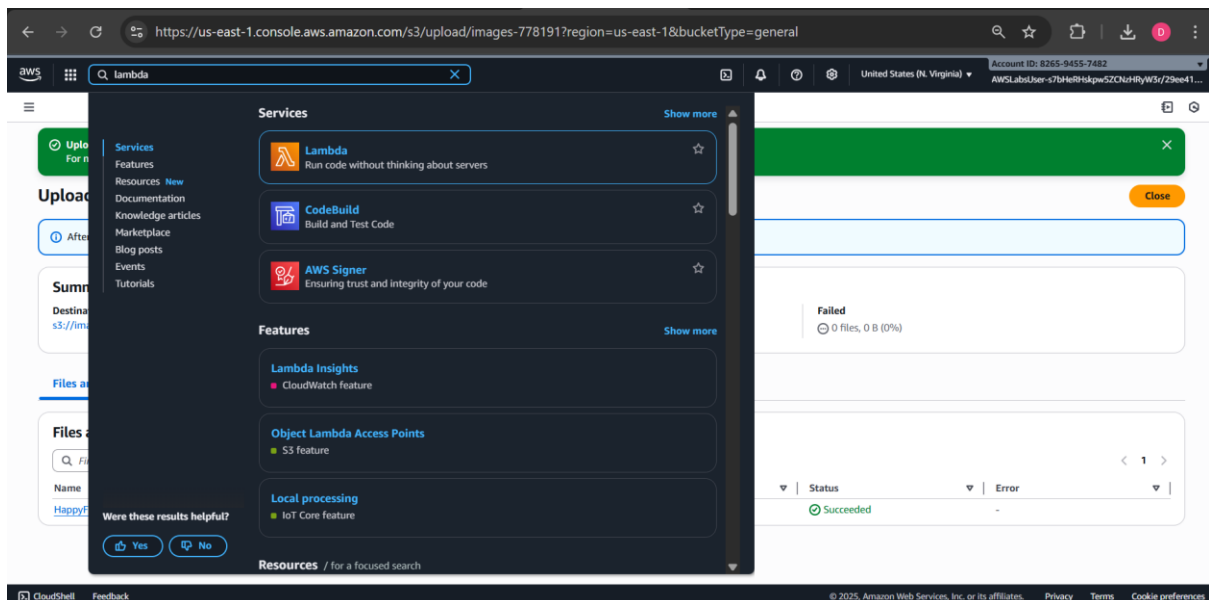


Task complete: You have created an S3 bucket and stored an object on it.

Task 2: Create an AWS Lambda function

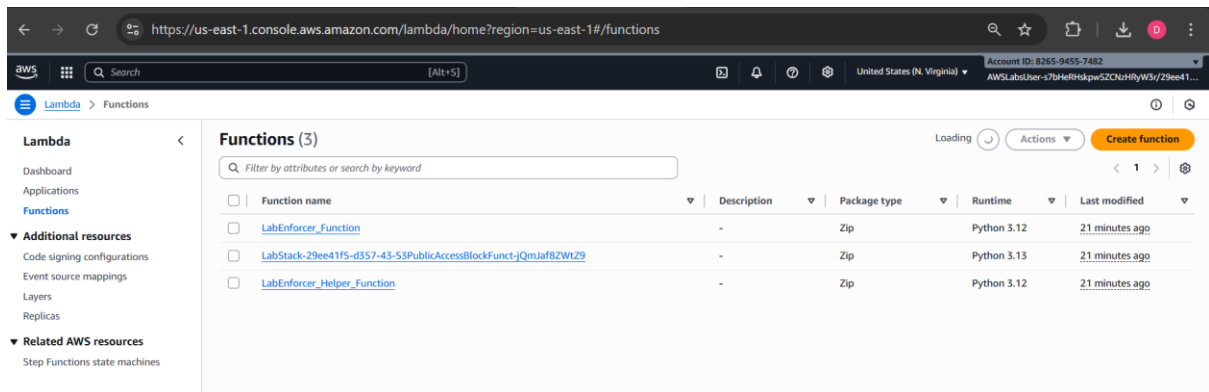
In this task, you create an AWS Lambda function that reads an image from Amazon S3, resizes the image, and then stores the new image in Amazon S3.

16. At the top of the AWS Management Console, in the unified search bar, search for and choose Lambda



17. Choose **Create function**.

Note: *Blueprints* are code templates for writing Lambda functions. Blueprints are provided for standard Lambda triggers such as creating Alexa skills and processing Amazon Kinesis Firehose streams. This lab provides you with a pre-written Lambda function, so you **Author from scratch**.

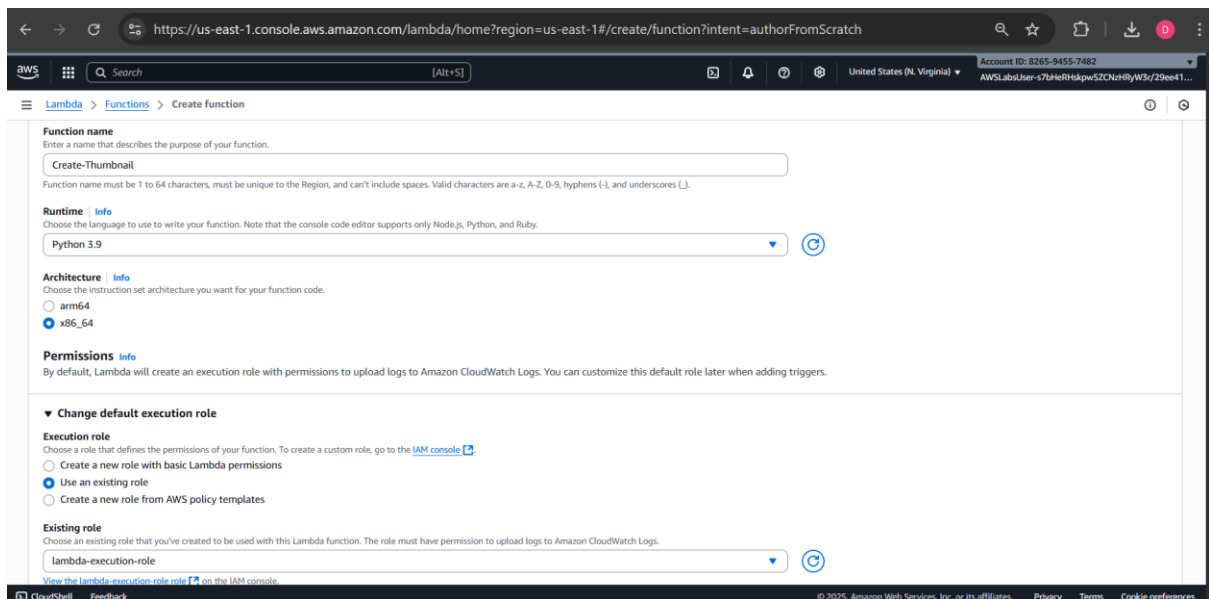


18. On the **Create function** page, select **Author from scratch**.

19. In the **Basic information** section:

- For **Function name**:, enter **Create-Thumbnail**
- For **Runtime**:, select **Python 3.9**.
- Expand **Change default execution role**.
- For **Execution role**:, choose **Use an existing role**.
- For **Existing role**:, select **lambda-execution-role**.

Note: This *role* grants permission to the Lambda function to access Amazon S3 to read and write the images.



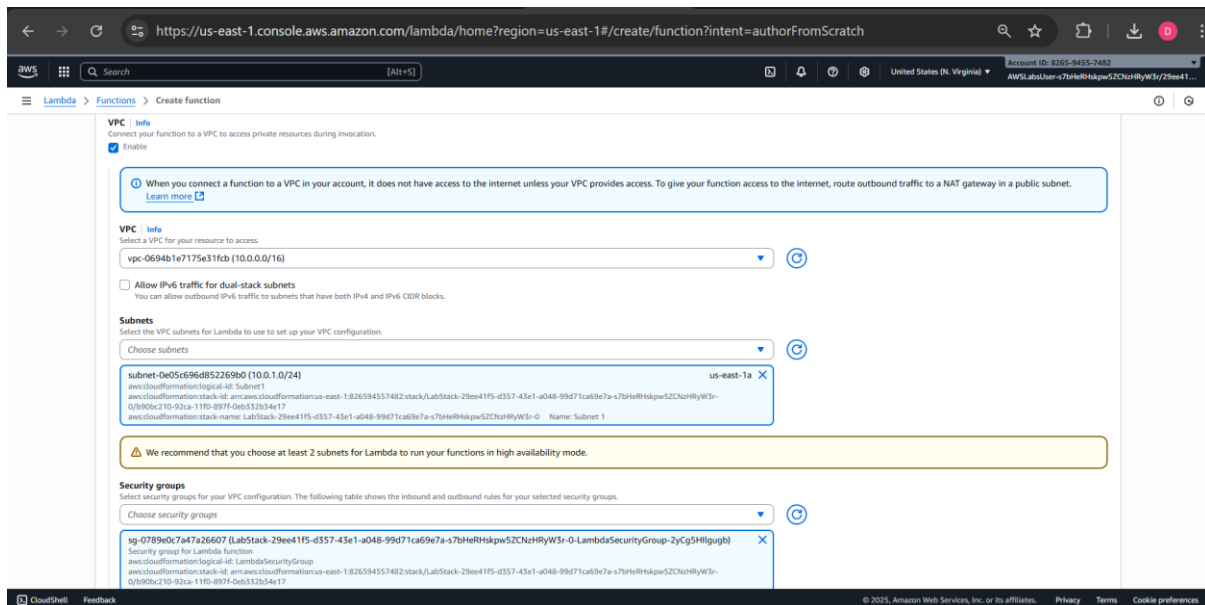
20. Expand the **Additional configurations** section.

Note: To make this lab more secure, you must attach your function to a VPC, subnet, and security group that have been created for you. Conventionally, Lambda functions do not need to be attached to a VPC, but this lab will not let you create the function without the addition of defined networking.

21. In the **Additional configurations** section:

- Choose **VPC**.

- For **VPC**, select the VPC with CIDR range **10.0.0.0/16**.
- For **Subnets**, select the subnet with CIDR range **10.0.1.0/24**.
- For **Security groups**, select the security group with **LambdaSecurityGroup** in its name.



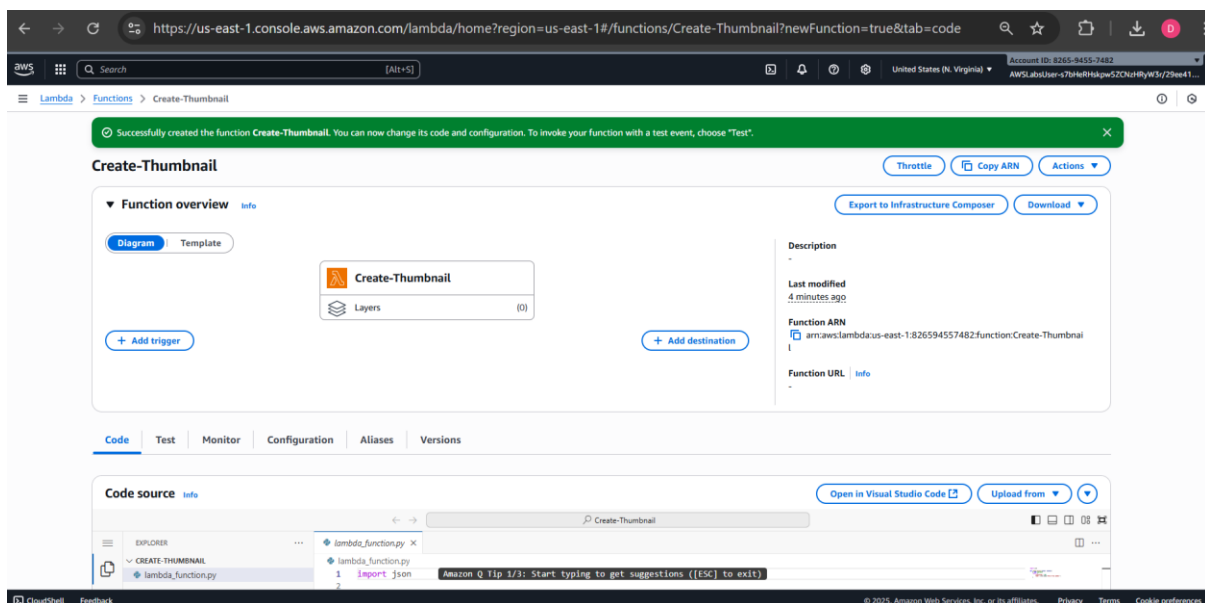
22. Choose **Create function** .

23. On the **Getting started** pop-up, choose **Dismiss** .

A page is displayed with your function configuration.

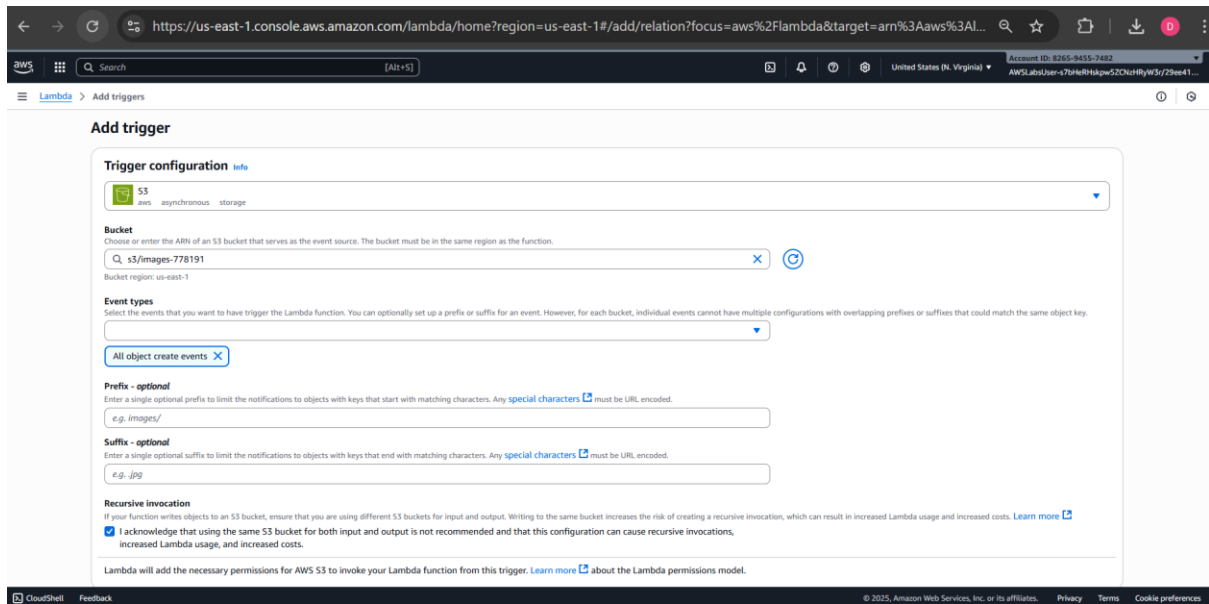
Note: Creating a VPC-attached Lambda function takes a few minutes. Wait until a banner at the top of the page displays **Successfully created the function Create-Thumbnail**. to proceed.

AWS Lambda functions can be **triggered** automatically by activities such as data being received by Amazon Kinesis or data being updated in an Amazon DynamoDB database. For this lab, you trigger the Lambda function whenever a new object is created in your Amazon S3 bucket.

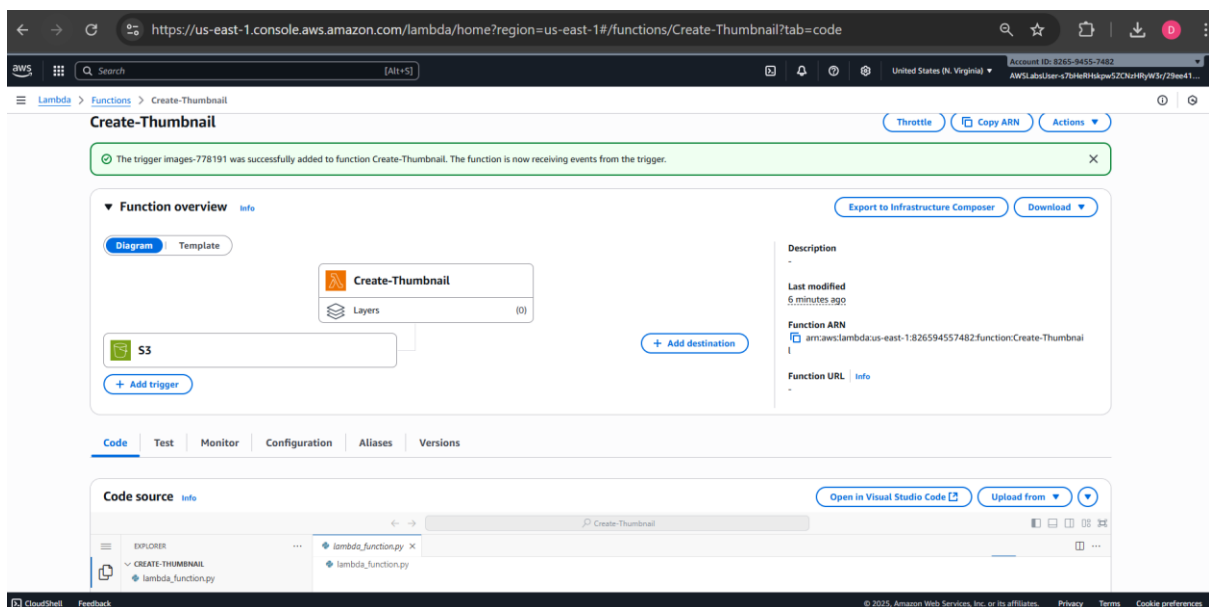


24. Choose **Add trigger** .

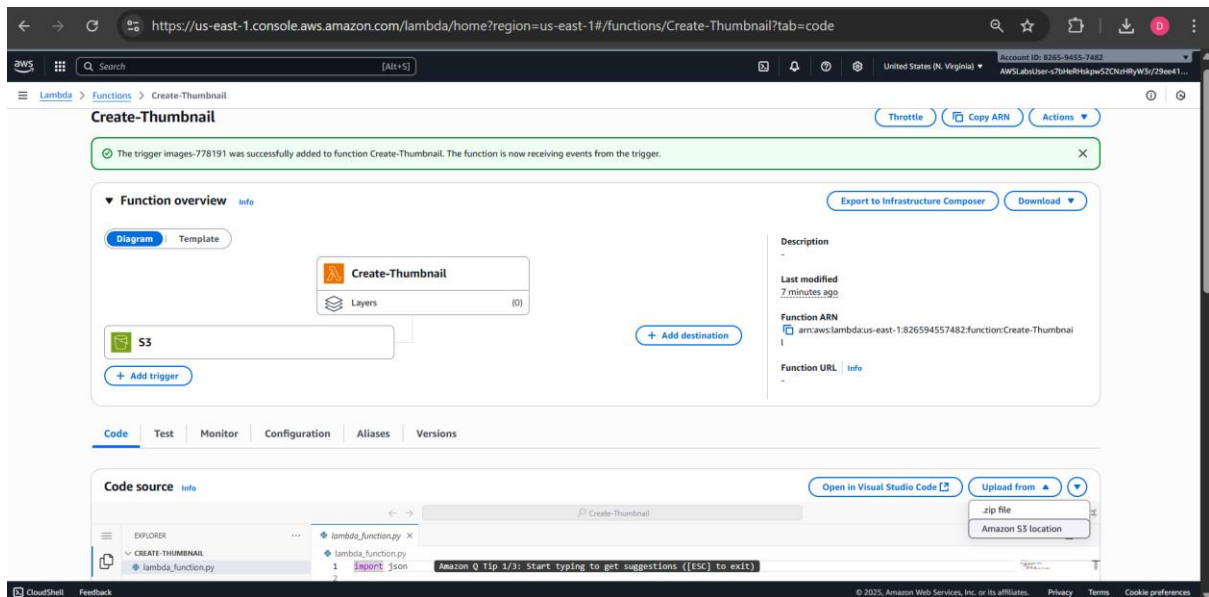
25. For **Trigger configuration**, use the drop-down selector to select **S3**



28. Choose the **Code** tab.



29. Choose **Upload from** and select **Amazon S3 location**.



30. For **Amazon S3 link URL**;, copy and paste the value for **AmazonS3LinkURL** from the panel to the left of these instructions.

31. Choose **Save** .

Note: The *CreateThumbnail.zip* file contains the following Lambda function: Do not copy this code – it is just showing you what is in the Zip file.

```
import boto3
```

```
import os
```

```
import sys
```

```
import uuid
```

```
from PIL import Image
```

```
import PIL.Image
```

```
s3_client = boto3.client('s3')
```

```
def resize_image(image_path, resized_path):
```

```
    with Image.open(image_path) as image:
```

```
        image.thumbnail((128, 128))
```

```
        image.save(resized_path)
```

```
def handler(event, context):
```

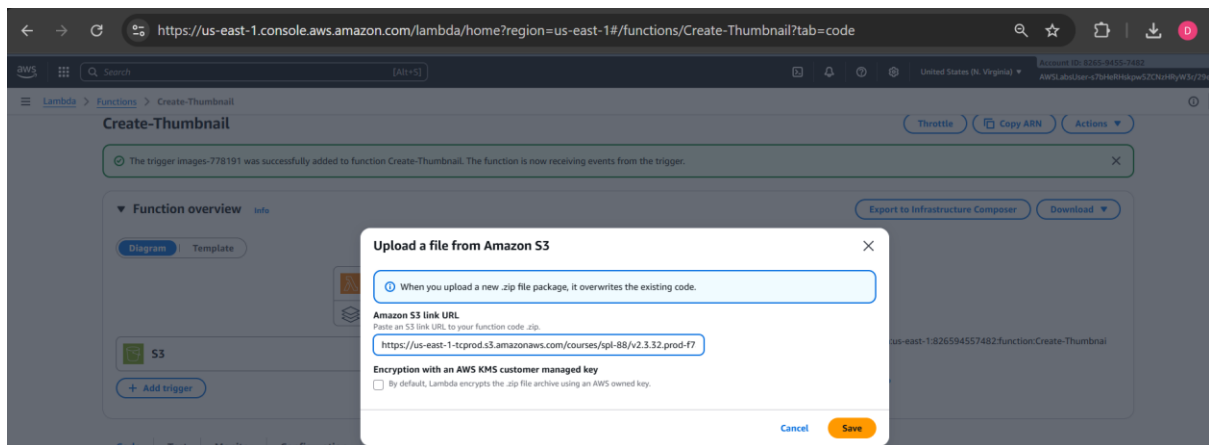
```
    for record in event['Records']:
```

```

bucket = record['s3']['bucket']['name']
key = record['s3']['object']['key']
download_path = '/tmp/{}'.format(uuid.uuid4(), key)
upload_path = '/tmp/resized-{}'.format(key)

s3_client.download_file(bucket, key, download_path)
resize_image(download_path, upload_path)
s3_client.upload_file(upload_path, '{}-resized'.format(bucket), key)

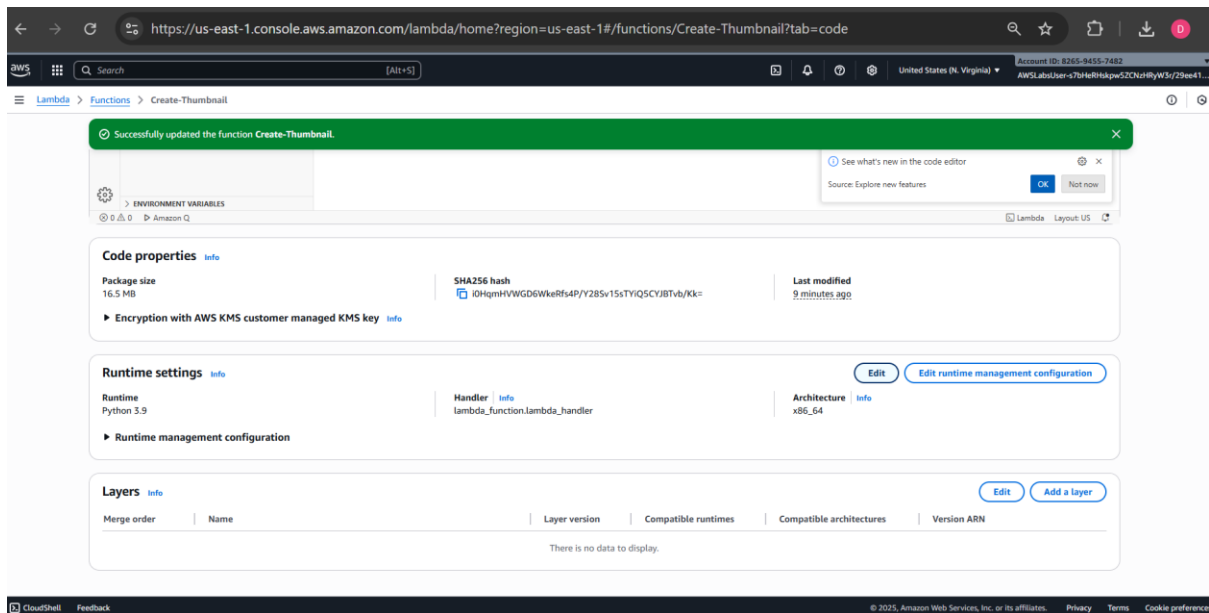
```



32. **(Optional step):** Examine the above code. It is performing the following steps:

- Receives an Event, which contains the name of the incoming object (Bucket, Key)
- Downloads the image to local storage
- Resizes the image using the *Pillow* library
- Uploads the resized image to the *-resized* bucket

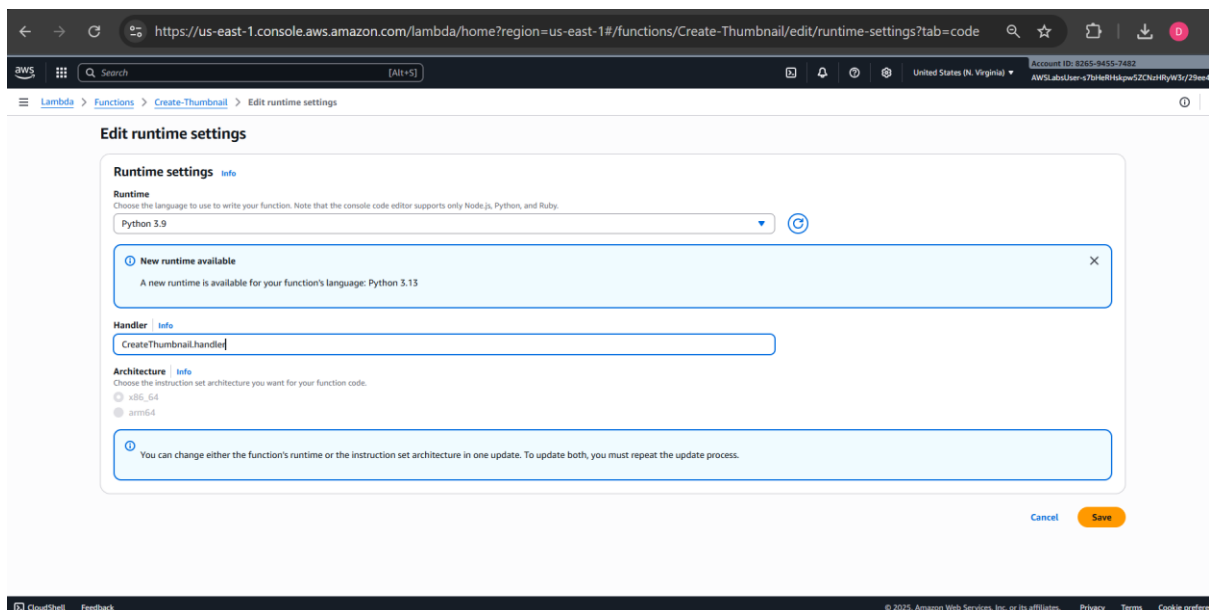
33. In the **Runtime settings** section, choose **Edit** .



34. For **Handler**, replace the existing text with `CreateThumbnail.handler`

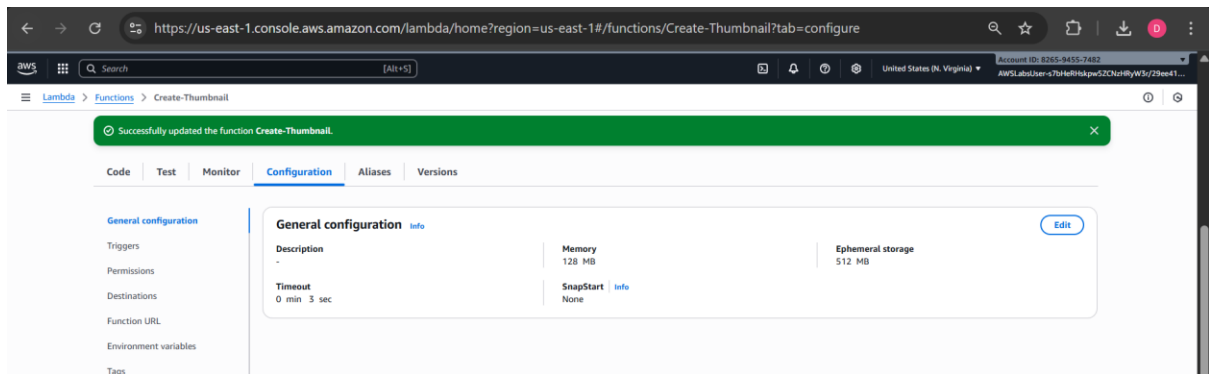
35. Choose **Save** .

Note: Make sure you set the **Handler** field to the above value, otherwise the Lambda function will not be found.



36. Choose the **Configuration** tab.

37. Choose **General configuration** on the left, then choose **Edit** .

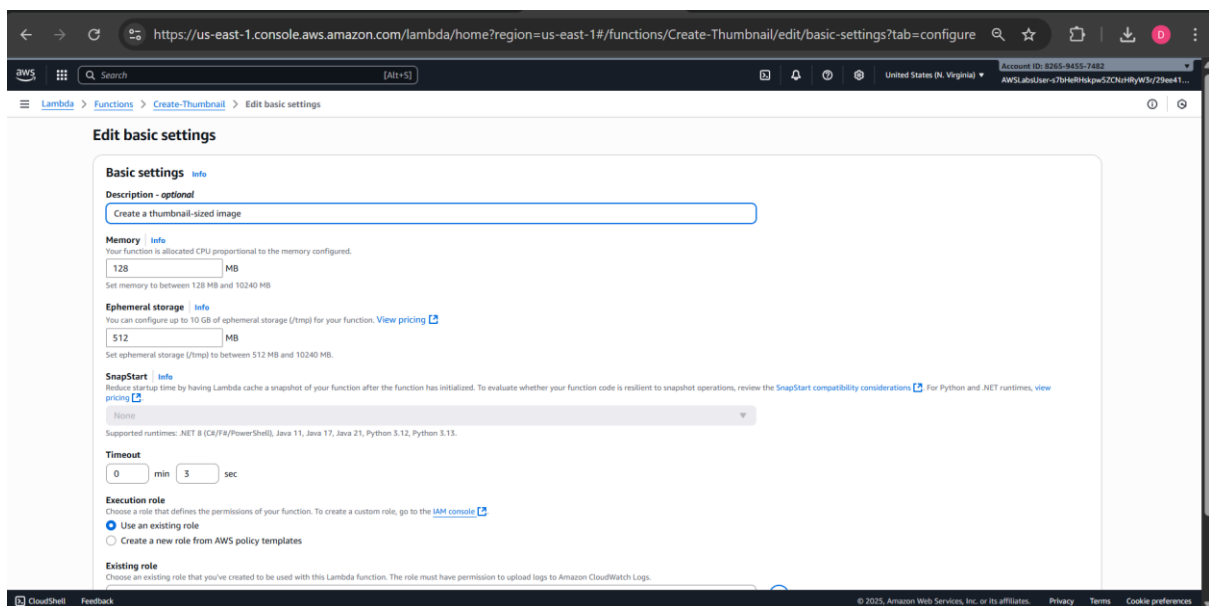


38. For **Description - optional**, enter Create a thumbnail-sized image

Note: Leave the other settings as default, but here is a brief explanation of these settings:

- **Memory** defines the resources that will be allocated to your function. Increasing memory also increases CPU allocated to the function.
- **Timeout** sets the maximum duration for function execution.

39. Choose **Save**.



Task complete: Your Lambda function has now been configured.

Task 3: Test your function

In this task, you test your Lambda function. This is done by simulating an event with the same information normally sent from Amazon S3 when a new object is uploaded.

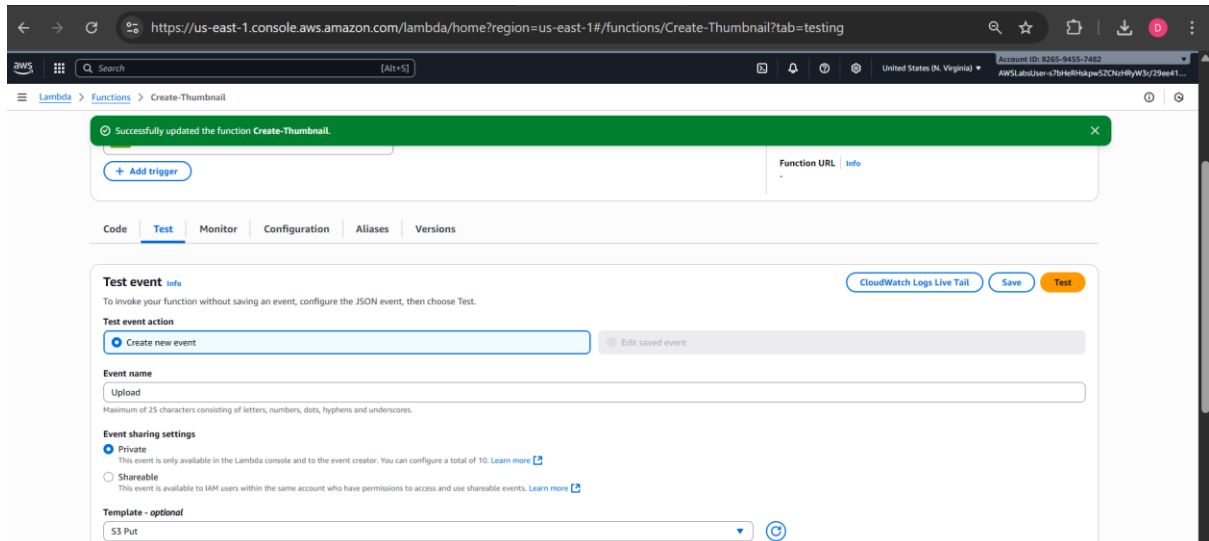
40. Choose the **Test** tab.

41. In the **Test event** section, select **Create new event** and then configure:

- For **Event name**, enter Upload

- For **Template - optional**:, select **S3 Put**

A sample template is displayed that shows the event data sent to a Lambda function when it is triggered by an upload into Amazon S3. You need to edit the bucket name so that it uses the bucket you created earlier.

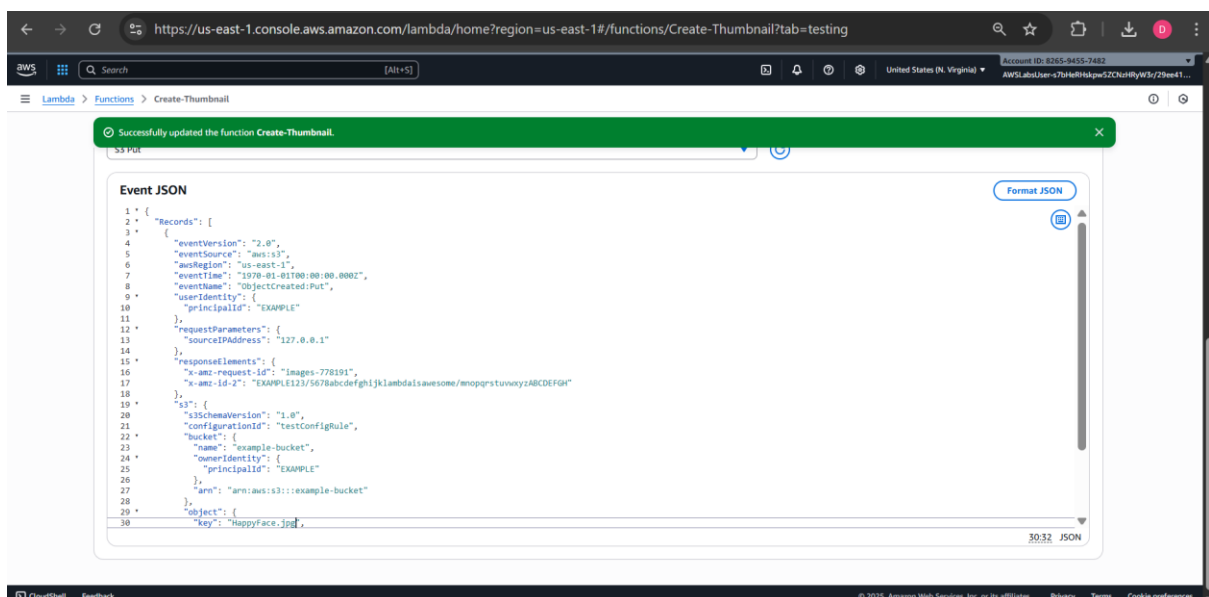


42. Replace **example-bucket** text in the **Event JSON** section with the name of your images bucket (e.g. *images-123456789*) that you copied to your text editor.

Note: The *example-bucket* text appears twice, as the bucket name and the bucket arn (replace only the example-bucket text, leave the rest of the arn intact). Be sure to replace it in both locations.

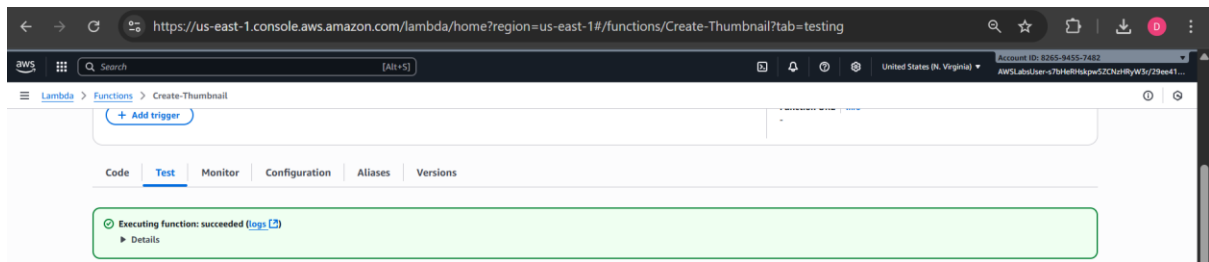
43. Replace **test%2Fkey** text in the **Event JSON** section with the name of the picture that you uploaded. This should be HappyFace.jpg

Note: this text is the object key in the JSON, near the bottom.



44. Choose **Test**.

After a moment, you should see the message: *Executing function: succeeded*

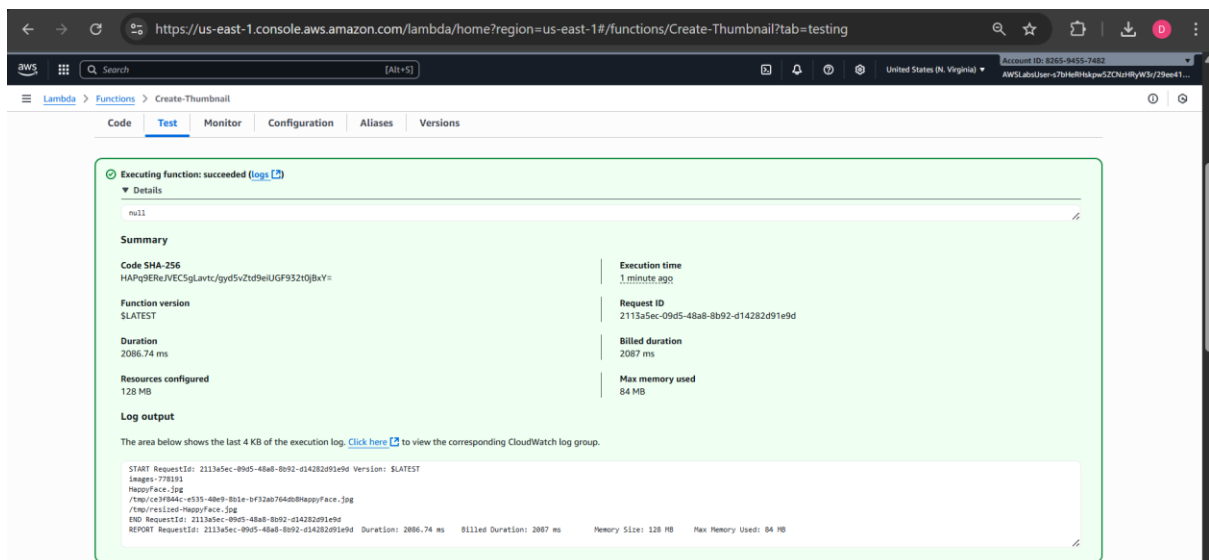


45. Choose **Details** to expand it.

You are shown information including:

- Execution time
- Resources configured
- Maximum memory used
- Log output

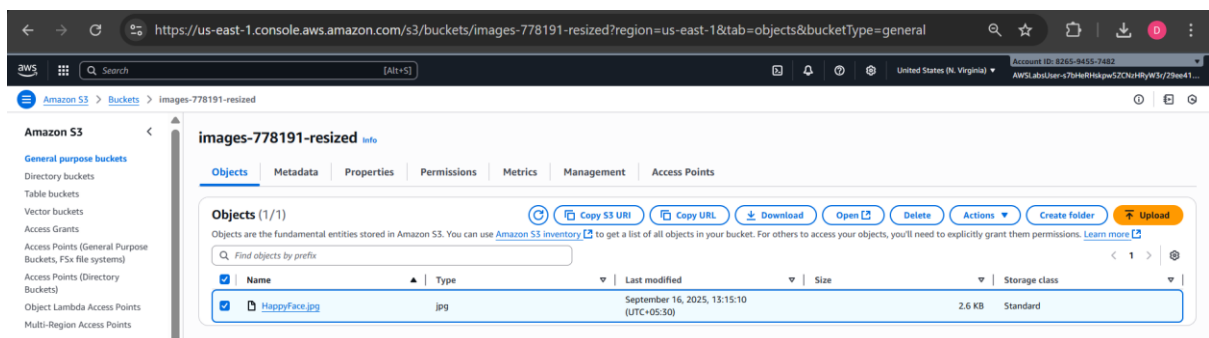
You can now view the resized image that was stored in Amazon S3.



46. At the top of the AWS Management Console, in the unified search bar, search for and choose S3

47. Choose the name of your **-resized** bucket (which is the second bucket you created).

48. Select the checkbox for **HappyFace.jpg**.

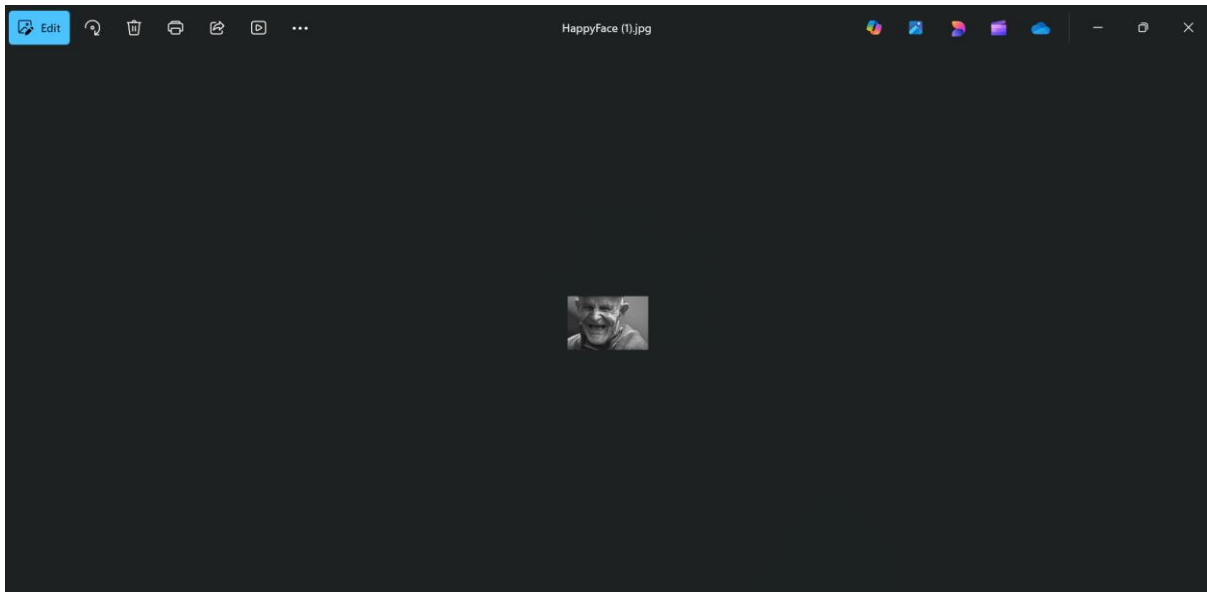


49. Choose **Open** .

Note: Depending on your browser and settings, this may either open the picture or download it (usually with a *(1)* added to the filename). If the browser only downloaded the file, open it on your machine using whatever app is appropriate.

The image should now be a smaller thumbnail of the original image.

You are welcome to upload your own images to the *images*- bucket and then check for thumbnails in the *-resized* bucket.



Task complete: You have triggered a Lambda function and observed the results.

Task 4: Monitoring and logging

In this task, you monitor AWS Lambda functions to identify problems and view log files to assist in debugging.

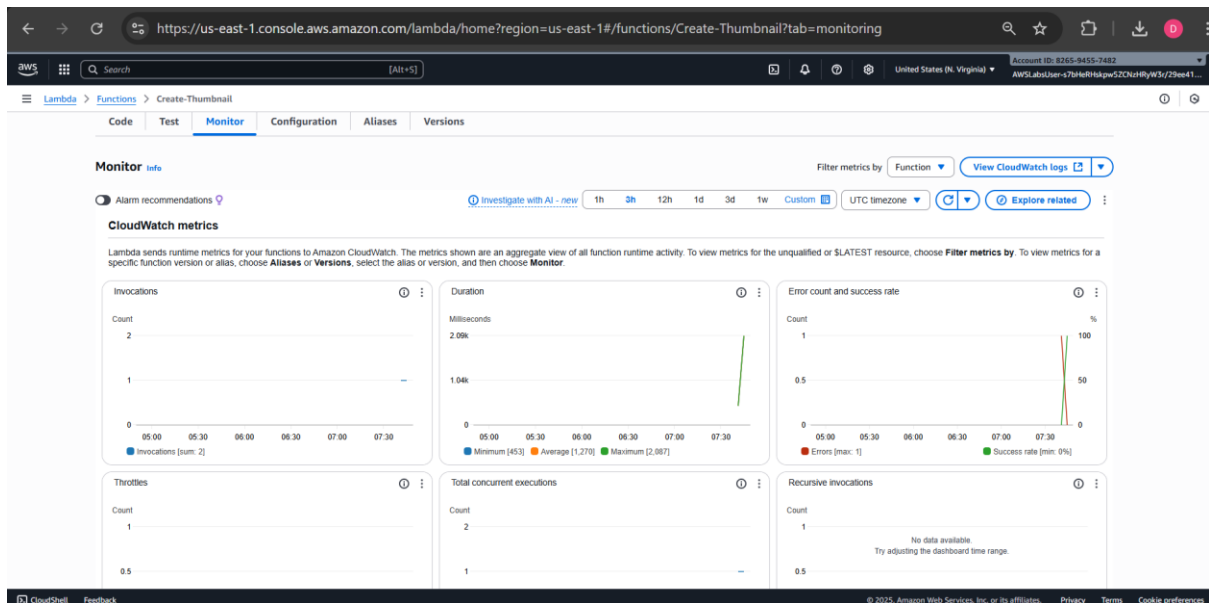
50. At the top of the AWS Management Console, in the unified search bar, search for and choose Lambda
51. Choose your **Create-Thumbnail** function (the link, not the checkbox).
52. Choose the **Monitor** tab.

The console displays graphs including:

- **Invocations:** The number of times that the function was invoked.
- **Duration:** The average, minimum, and maximum execution times.
- **Error count and success rate (%):** The number of errors and the percentage of executions that completed without error.

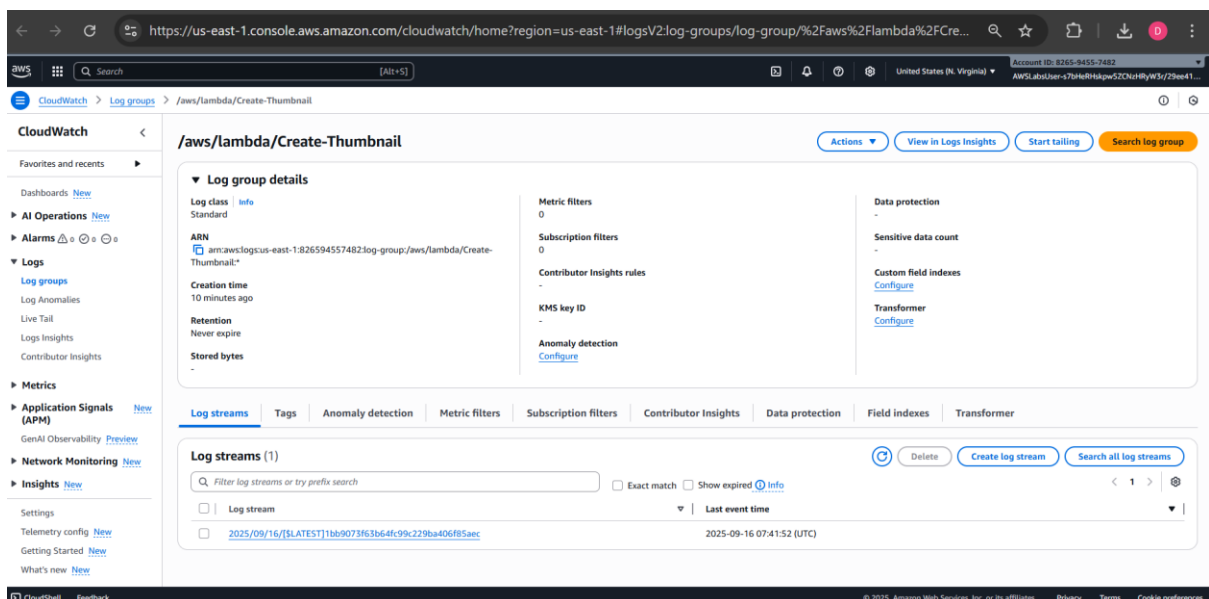
- **Throttles:** When too many functions are invoked simultaneously, they will be throttled. The default is 1000 concurrent executions.
- **Async delivery failures:** The number of errors that occurred when Lambda attempted to write to a destination or dead-letter queue.
- **Iterator Age:** Measures the age of the last record processed from streaming triggers (Amazon Kinesis and Amazon DynamoDB Streams).
- **Total concurrent executions:** The number of function instances that are processing events.

Log messages from Lambda functions are retained in **Amazon CloudWatch Logs**.



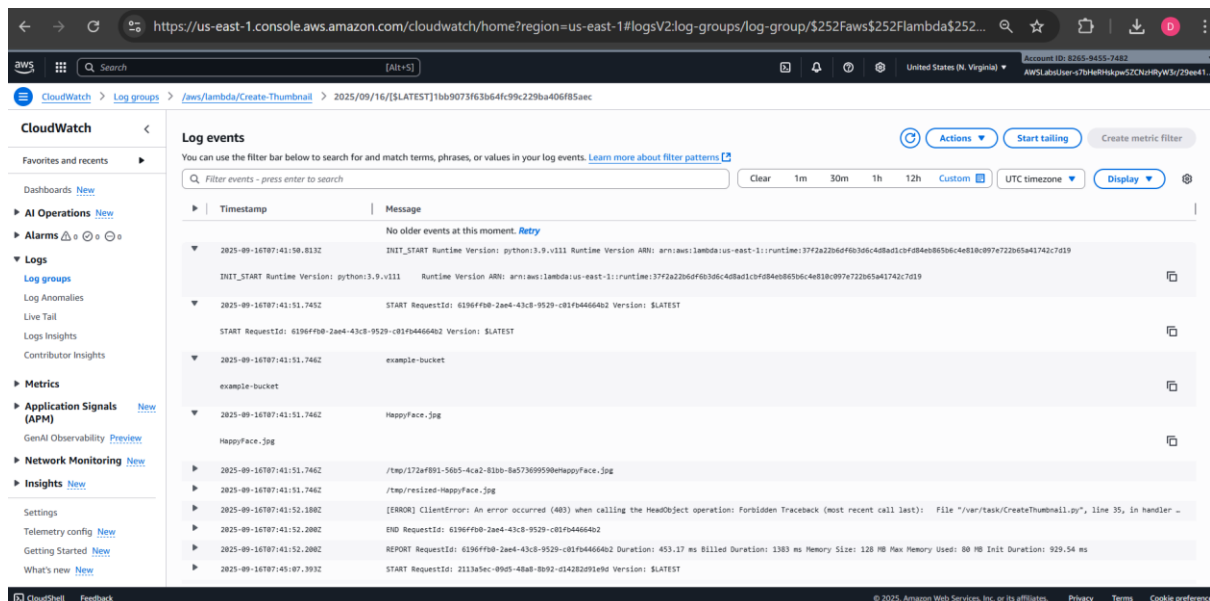
53. Choose **View CloudWatch logs**.

54. Choose the **Log Stream** that appears (the link, not the checkbox).



55. Expand each message to view the log message details.

The Event Data includes the Request Id, the duration (in milliseconds), the billed duration (rounded up to the nearest 100 ms, the Memory Size of the function and the Maximum Memory that the function used. In addition, any logging messages or print statements from the functions are displayed in the logs. This assists in debugging Lambda functions.



Task complete: You have monitored AWS Lambda to S3 functions using Amazon CloudWatch.

Conclusion

You have successfully done the following:

- Created an AWS Lambda function
- Configured an Amazon S3 bucket as a Lambda Event Source
- Triggered a Lambda function by uploading an object to Amazon S3
- Monitored AWS Lambda S3 functions through Amazon CloudWatch Log

End lab

Follow these steps to close the console and end your lab.

56. Return to the **AWS Management Console**.
57. At the upper-right corner of the page, choose **AWS Labs User**, and then choose **Sign out**.
58. Choose **End Lab** and then confirm that you want to end your lab.

Additional Resources

- [AWS Lambda documentation](#)